OsdagBridge Report Generator

Technical Architecture, Modifications & Visual Engineering Report

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1 Executive Summary

This report documents the technical architecture, code refactoring, and LaTeX visual rendering enhancements implemented on the **OsdagBridge Report Generator**.

The legacy reporting subsystem previously assembled PDF reports via procedural string-concatenation scripts (`chap1.py` through `chap8.py`), tightly coupling calculation outputs with raw LaTeX formatting. This approach caused margin overflows on multi-girder bridge configurations (> 5 girders), footer overlap issues, unformatted math strings, and scattered styling parameters.

The subsystem has been re-architected into a **10-Tier Semantic Report Engine**. The architecture decouples **Engineering Domain Facts**, **Document Abstract Syntax Tree (AST)**, **Centralized Styling (`theme.py`)**, and **LaTeX Rendering (`renderer.py`)**, backed by automated **PDF Preflight Layout Inspection** and a **350-test automated verification suite**.

2 10-Tier Semantic Architecture Pipeline

Tier	Component	Core Responsibility	Why Introduced (Rationale)
1	**Raw Payload**	Captures solver results, grillage data, and inputs.	Decouples raw solver data structures.
2	**Payload Validation**	Fails fast on invalid span ≤ 0, girders ≤ 1, or missing loads.	Prevents silent crashes on malformed models.
3	**Provenance Tracking**	Records source dictionary keys, raw values, and unit conversions.	End-to-end data lineage; prevents unit errors.
4	**ReportFacts**	Immutable, strongly-typed Python domain dataclasses.	Eliminates data mutation; preserves None $\neq$ 0.0.
5	**Fact Validation**	Asserts positive dimensions, finite demands, non-negative capacity.	Enforces domain invariants before AST building.
6	**ReportDocument AST**	In-memory AST (`Chapter`, `Section`, `Table`, `TableGroup`, `Chart`).	Completely separates content from LaTeX markup.
7	**Structural Manifest**	Computes AST fingerprints and asserts structural parity.	Tripwire preventing accidental check omissions.
8	**ReportTheme**	Single source of truth for page geometry, colors, table styles.	Eliminates scattered styling; instant global updates.
9	**LatexRenderer**	Emits `longtable` headers, `\cline` rules, and landscape flow.	Emits clean, robust LaTeX without ad-hoc strings.
10	**PDFPreflight Gate**	PyMuPDF binary scanner checking footer clearance and margins.	Catches physical page overflows post-compilation.

3 Scope of Work & Engineering Enhancements

Category	Requirement / Scope Item	Status	Key Technical Delivery
Core Requirement 1	Repeated Table Headers	**Completed**	4-tier `longtable` protocol (`\endfirsthead`, `\endhead` with *(con- tinued)*).
Core Requirement 2	Layout & Footer Overlap Fixes	**Completed**	Dynamic `\Needspace` clearance buffers + PyMuPDF automatec footer check.
Core Requirement 3	Load & Geometry Refactoring	**Completed**	Separated vehicle live loads, footway, wind, seismic, and ther- mal tables.
Core Requirement 4	Utilization Ratio Visualizations	**Completed**	**Figure 5.1** summary chart with UR = 1.0 red dashed code threshold line.
Core Requirement 5	Material Take-off Bar Charts	**Completed**	**Figures 7.1, 7.2, 7.3** for Steel, Concrete, and Rebar with sequential numbering.
Core Requirement 6	Centralized Theme System	**Completed**	Dedicated `theme.py` acting as single source of truth for all lay- out/color rules.
Additional Enhancement	Multi-Girder Scalability	**Enhanced**	Verticalized Table 1 (5 columns), scaling gracefully across 3, 5 9, 10+ girders.
Additional Enhancement	Grouped Table Aesthetics	**Enhanced**	Subtle `\cline{2-N}` inner parameter dividers with full `\hline` group boundaries.
Additional Enhancement	LaTeX Math AST Protection	**Enhanced**	AST `Math(...)` token isolation eliminating ¿ font artifacts and raw TeX leaks.
Additional Enhancement	Preflight Safety Tripwires	**Enhanced**	Post-compilation automated physical PDF margin and footer collision detection.

4 Specific Files Altered & Created Under Report Generator

File Path	Status	Primary Purpose & Responsibility
`src/osdagbridge/core/report_engine/theme.py`	**NEW**	Centralized styling configuration (`PageGeometry`, `Typogra` `ColorPalette`, `TableStyle`, `ChartStyle`).
`src/osdagbridge/core/report_engine/document.py`	**NEW**	Intermediate AST definitions (`ReportDocument`, `Chapter`, `Secti` `TableGroup`, `Column`, `Chart`).
`src/osdagbridge/core/report_engine/renderer.py`	**NEW**	Production LaTeX rendering engine, math escaping, `longtable` ment, and landscape section flow.
`src/osdagbridge/core/report_engine/chart_generators.py`	**NEW**	Matplotlib chart generation service bound exclusively to `Report` ors and typography.
`src/osdagbridge/core/report_engine/preflight.py`	**NEW**	Automated PyMuPDF (`fitz`) binary PDF inspector for footer and horizontal margin verification.
`src/osdagbridge/core/report_engine/manifest.py`	**NEW**	Structural manifest extraction and legacy vs. semantic parity v engine.
`src/osdagbridge/core/report_engine/facts/`	**NEW**	Typed domain fact models (`inputs.py`, `loads.py`, `design_c` `material_takeoff.py`).
`src/osdagbridge/core/report_engine/validation.py`	**NEW**	Input payload and domain fact invariant validators.
`src/osdagbridge/core/report_engine/provenance.py`	**NEW**	`ProvenanceTracker` recording raw dictionary origins and regis transformations.
`src/osdagbridge/core/reports/report_generator.py`	**MODIFIED**	Main orchestrator; integrated semantic engine, cleaned runnin and linked preflight checks.
`src/osdagbridge/core/reports/executive_summary.py`	**MODIFIED**	Re-architected Table 1 into a vertical 5-column structure (`Girder ID | Section | Check | UR`).
`src/osdagbridge/core/reports/chap6.py`	**MODIFIED**	Constrained Figure 6.1 height to `10.5cm, keepaspectratio` to view on opening page.

5 Centralized Theme Architecture (`theme.py`)

The centralized formatting configuration module (`theme.py`) serves as the **single source of truth** for all visual properties:

```
ReportTheme (Immutable Single Source of Truth)
 page: PageGeometry          → A4 dimensions, 20mm margins, 25mm footer reserve
 typography: TypographyStyle  → 11pt Computer Modern body, 14pt headings, bold captions
 colors: ColorPalette         → Osdag Brand Green (#91B014), Deep Slate Navy (#1E293B), Neutral Gray (#94A3B8)
 table_styles: TableStyle     → 6pt padding, inner_group_rule="subtle" (\cline), boundary_rule="strong" (\hline)
 charts: ChartStyle           → 14.5cm x 5.5cm dimensions, 300 DPI, sans-serif typography
```

6 Key Technical Changes: Before vs. After Comparison

Report Element	Legacy Implementation (Before)	Semantic Report Engine (After)
Executive Summary Table 1	Horizontal columns ($G_1 \ldots G_{10}$) causing right-margin overflow.	Vertical 5-column table (`Girder` | `Member` | `Section` | `Check` | `UR`), perfectly scalable.
Multi-Page Tables	Headers omitted on continuation pages; data disconnected across breaks.	Automatic `\endhead` repeated headers with *(continued)* captions on all pages.
Grouped Girder Tables	No internal dividing lines between parameter rows, resulting in cluttered blocks.	Subtle `\cline{2-N}` internal rules with full-width `\hline` group boundaries.
Charts & Visualizations	Default Matplotlib rainbow palette (blue, orange, teal); missing figure numbers.	Unified Osdag Brand Green (`#91B014`), explicit `N/A` badges, and auto figure numbers.
LaTeX Math Formatting	Raw string escaping exposed unformatted code (`\times`, `\leq`, `\phi`) and ¿ artifacts.	Tokenized math isolation with AST `Math(...)` objects preserving mathematical typography.
Landscape Summary Flow	Default vertical stretching caused portrait page footers to float mid-page.	Isolated `osdaglandscape` environments with `\raggedbottom` and full-width header rules.
Quality Verification	Manual visual inspection of compiled PDFs.	Automated PyMuPDF `PDFPreflight` checking bounding boxes for margin/footer collisions.

7 Verification & Production Acceptance

Configuration Matrix	Girder Count	Page Count	Footer Collision	Margin Overflow
Configuration A	3 Girders	44 Pages	**PASS**	**PASS**
Configuration B	5 Girders	51 Pages	**PASS**	**PASS**
Configuration C	10 Girders	69 Pages	**PASS**	**PASS**
Configuration D (Partial)	4 Girders	44 Pages	**PASS**	**PASS**
10-Girder Production Report	10 Girders	78 Pages	**PASS**	**PASS**

Automated Test Suite Status: 350 PASSED, 1 SKIPPED, 0 FAILURES (Execution time: 13.02s).

- **Contract & Invariant Tests:** 11/11 passed (< 0.4s). | **Structural Manifest Parity:** 6/6 passed, 0 regressions.
- **Data Provenance & Lineage:** 7/7 passed. | **Multi-Configuration Acceptance:** Certified on real PDFs.

8 Known Limitations & Non-Goals

1. **Solver Boundary:** The Report Engine formats and validates data produced by OsdagBridge; it does not replace the grillage solver.
2. **Preflight Scope:** `PDFPreflight` verifies document layout and geometry; it is not an independent mathematical re-solver.
3. **Compilation Backend:** Generates standardized LaTeX and utilizes `pdflatex` as the production compilation backend.
4. **Optional Components:** Figures and sections remain configuration-dependent based on user UI selections.

9 Submission Deliverables

Deliverable	Description & Contents
`Report_Before.pdf`	Original 10-girder baseline report generated prior to refactoring, illustrating legacy issues (Table 1 overflow, missing continuation headers).
`Report_After.pdf`	Final production 10-girder report with vertical Table 1, repeated headers, subtle `\cline` separators, and certified preflight clearance.
`Technical_Changes_Report.pdf`	Standalone compiled technical architecture report detailing code modifications, `ReportTheme`, and design rationales.
`README.md`	Submission manifest, repository link, collaborator info, and test validation summary.

10 Conclusion

The OsdagBridge Semantic Report Engine has completed the defined production acceptance and regression verification process and is frozen at Git tag `report-engine-production-accepted`. The subsystem is robust, maintainable, scalable across all girder configurations, and fully certified for production release.